

Immersive Health Data Visualization in Virtual Reality

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ABSTRACT

During the recent pandemic, critical decisions were taken by public healthcare officials by viewing and analyzing data. Doing so in an immersive environment like virtual reality (VR) would have provided more context and aided with engagement and better performance. As a solution to this problem, an interactive 3D virtual reality data visualization tool is being developed by a team of Unreal developers, public-health domain experts, and user experience (UX) researchers at the Imaging Research Center, UMBC. This research aims to understand the interaction mechanisms that a novice user from the public health domain might face when interacting with an immersive data visualization tool. The results show that public policy and emergency health services workers found it beneficial to interact and find relationships between data in an immersive VR environment and were able to offer design recommendations for a more effective and engaging experience.

Index Terms: Human-centered computing—Immersive analytics—Virtual reality—Public health data visualization

1 INTRODUCTION AND MOTIVATION

Virtual reality (VR) has excellent potential for visualizing large amounts of data due to the unlimited space, natural interaction, low task load, high user engagement and enhanced user performance. Donalek et al. [1] describe how technologies associated with immersive VR are becoming more powerful, more ubiquitous, and affordable, and how they open novel ways of interaction and collaboration. By overlaying georeferenced data on a map in an immersive VR environment, Moran et al. [4] achieved situational awareness and cognition. The emerging field of immersive analytics (IA) investigates how immersive Augmented Reality (AR) and VR technologies support data analysis and decision-making [2, 8]. Marriott et al. [2] posit that improved displays, the intuitive use of the third dimension, and the freedom to work beyond the monitor offer unique opportunities for data visualization in IA. Previous research has shown that immersive data visualization can enhance user performance. Immersive Environments (IEs) create the impression of a user being “present” in the visualization, resulting in a data representation that is natural to the viewer and requires less cognitive strain [6]. This leads to amplified cognition and increased user acceptance and performance in tasks typical for information exploration [9]. Millais et al. [3] found no overall task-workload difference between traditional visualizations and visualizations in VR. At the same time, they found improved accuracy and depth of insights that users gain in VR. VR also made participants feel more satisfied and successful when exploring data. Raja et al. [5] found that the immersive environment did not present higher task loads and a combination of a high degree of physical immersion and head tracking resulted in lower task times and disorientation and higher usefulness ratings than 2D displays with or without head tracking and 3D immersive displays with no head tracking. IEs are closer to natural interaction than many other

computer systems and inherently overcome the problems of interacting with 3D data. As the viewer is immersed in the environment and can almost “touch” it, more complex forms of interaction such as selection may also be provided intuitively and naturally to the viewer [6]. While body-based interactions seem a natural choice, Santos-Torres et al. [7] found that device-based interaction for 3D immersive maps enabled better performance.

This research sets out to examine what data display and interaction methods best support the user experience of public health decision-makers and what analytical behaviors emerge in immersive data visualization. The data domain selected for this research is public health since America’s public health system is marked by disparities driven by systemic inequalities as evidenced by the under-resourced and flawed public health response to COVID-19. Policymakers and healthcare practitioners would benefit from visualizing public health data in VR, as they currently lack the tools and practices to see, connect, synthesize, and convey the mechanisms of georeferenced data that provide more context. To offer a solution to this problem, an interactive 3D virtual reality data visualization tool is being developed by a team of Unreal developers, public-health domain experts and UX researchers at the Imaging Research Center, UMBC.

2 RESEARCH GOALS

My Ph.D. research is motivated by the challenges a novice user from the public health domain might face when interacting with this immersive data visualization tool. This work hopes to inform both the development of the visualization tool as well as answer the following research questions:

RQ1 How to display geographically referenced quantitative data in immersive 3D VR visualization?

RQ2 What interaction mechanisms best support public health data users in an immersive visualization space?

RQ3 What visual analytical behaviors can be supported through immersive, 3D VR visualization in the public health domain?

The visualization tool contains a map-based bar chart visualization of public health data as the central component of a large, multi-modal immersive data analysis environment where the goal is to help health care experts view, analyze and determine solutions to tackle different kinds of challenges the state of Maryland (MD), United States faces (see Figure 1). For example, determining the current prevalence of opioid use in MD and the incidence of opioid overdose, the public health interventions available to combat the opioid crisis, the relative impact of deploying a given intervention in each jurisdiction and the future allocation of resources toward a given intervention in each jurisdiction.

3 METHODS AND RESULTS

The adopted research approach is lab-based studies combining qualitative and quantitative data collected. The methods used so far include pre- and post-surveys (task load, user satisfaction, presence, motion sickness), participant observation, and interviews. Two user studies have been conducted: 1) a pilot mixed methods menu placement preference study with six UMBC graduate students to compare two menu locations in the immersive VR space and the ways it impacts data exploration and analysis and 2) a user study with ten

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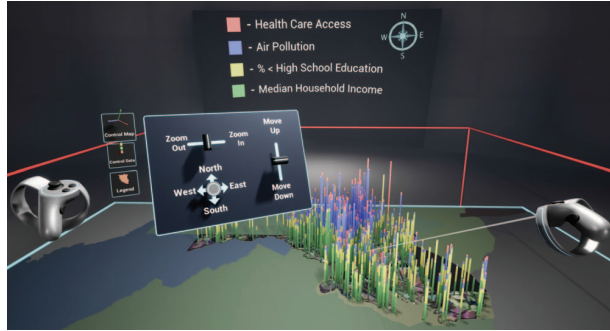


Figure 1: Immersive data visualization tool displaying health census data on the map of Maryland

public health policy and emergency health services (EHS) graduate students to understand the interaction challenges and potential benefits of using the data visualization tool. Study 1 compared two menus, one that is fixed to the wall of the VR environment and another one that is controller-based and pops up on the left hand of the user. The purpose of the study was to inform the design of the tool, which was still in its initial development phase, and provide the rationale to RQ1 regarding where to place the menu and why. The results suggest that most users preferred the menu to be constantly available on the wall with few design modifications such as having the option to resize the menu and move it anywhere in the space. Study 2 was domain-specific, where we recruited graduate students from the public policy program and EHS department at UMBC. As main tasks, the participants were given realistic health data comparison scenarios which helped them relate to their day-to-day data analysis and how a tool like that could change their whole experience. The results of study 2 inform RQ2 and RQ3 (partially). The applicability of the tool to other health domains, training methods and intensity for first-time VR users of this tool, walking in this environment, and potential design modifications were all gathered as part of the results of this study.

All the participants found it beneficial to interact and view data in this environment and could effectively understand the relative distributions of the variables in the different geographic areas. They offered suggestions for the future that they think would enhance the usability of the current system. Many of them suggested adding the ability to interact with the bar and get more detailed views of the data represented by the selected bar, such as numbers displayed near the selected bar. Another potential avenue of development was to select data by the map, such as selecting certain geographic areas and displaying data on only those to help avoid distraction when comparing specific areas. Other proposed functions included map rotation and the option to toggle the dominant hand to switch between hands. Lastly, the participants thought that factors such as VR technology being expensive, steep learning curve, possible motion sickness, walking disability, limited UX testing, and unavailability of bandwidth could be reasons to hinder the adoption of the visualization tool.

4 CONCLUSION

My research focus is at the intersection of human-computer interaction, virtual reality, immersive analytics, and information visualization. With my research, I aim to inform the design decisions of an immersive data visualization tool as well as identify generalized patterns to keep in mind when designing an immersive visualization tool for public health domain personnel. In the remainder of this Ph.D., further investigation will be carried out to understand public health officials' daily interaction activities, which will help

determine how such visualization and analytical behaviors could be supported in an immersive VR environment. Some areas that will be valuable to discuss and improve are:

- Narrow down existing research questions to address the needs of public health officials and present results accordingly.
- Find answers to some of the methodology questions such as what other measures to collect and analyze as part of the domain-specific user studies.

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